# Lecture 8

## EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

Associate Professor Lee Peng Hin
Tel: +65 6790 4474 | Email: ephlee@ntu.edu.sg

# Learning Objectives

By the end of this lecture, you should be able to:

- Describe the equivalent circuit of a DC motor.
- Identify the types of DC motors.
- Derive the torque-speed characteristics of a DC motor.
- Describe the losses and calculate the motor efficiency with given operating condition from the motor equivalent circuit.

# Introduction to DC Motors

- DC motors are DC machines used as motors.
- The same DC machine can operate as either a motor or a generator.
- DC power systems are still common in cars, trucks and aircrafts. When a vehicle has a DC power system, it makes sense to consider using DC motors.
- Even if no DC power sources are available, solid state rectifier and chopper circuits are used to create the necessary DC power for the DC motors.
- In areas where wide variations in torque-speed characteristics are required, such as in steel mills, DC motors are needed.

# Speed Regulation of DC Motors

- DC motors are compared by their speed regulations which is defined by:

$$\mathrm{SR} = \frac{\omega_{m,nl} - \omega_{m,fl}}{\omega_{m,fl}} \times 100\% = \frac{n_{m,nl} - n_{m,fl}}{n_{m,fl}} \times 100\%$$

- It is the change in speed from no load to full load, expressed as a percentage of full-load speed.

- The magnitude of the $\mathrm{SR}$ gives an indication of how steep the slope of the torque-speed curve is.

# Example 1

A 120-V, 1750 r/min, 5-hp motor, operating at rated conditions, has a speed regulation of 4.0 percent. Determine the no-load speed.

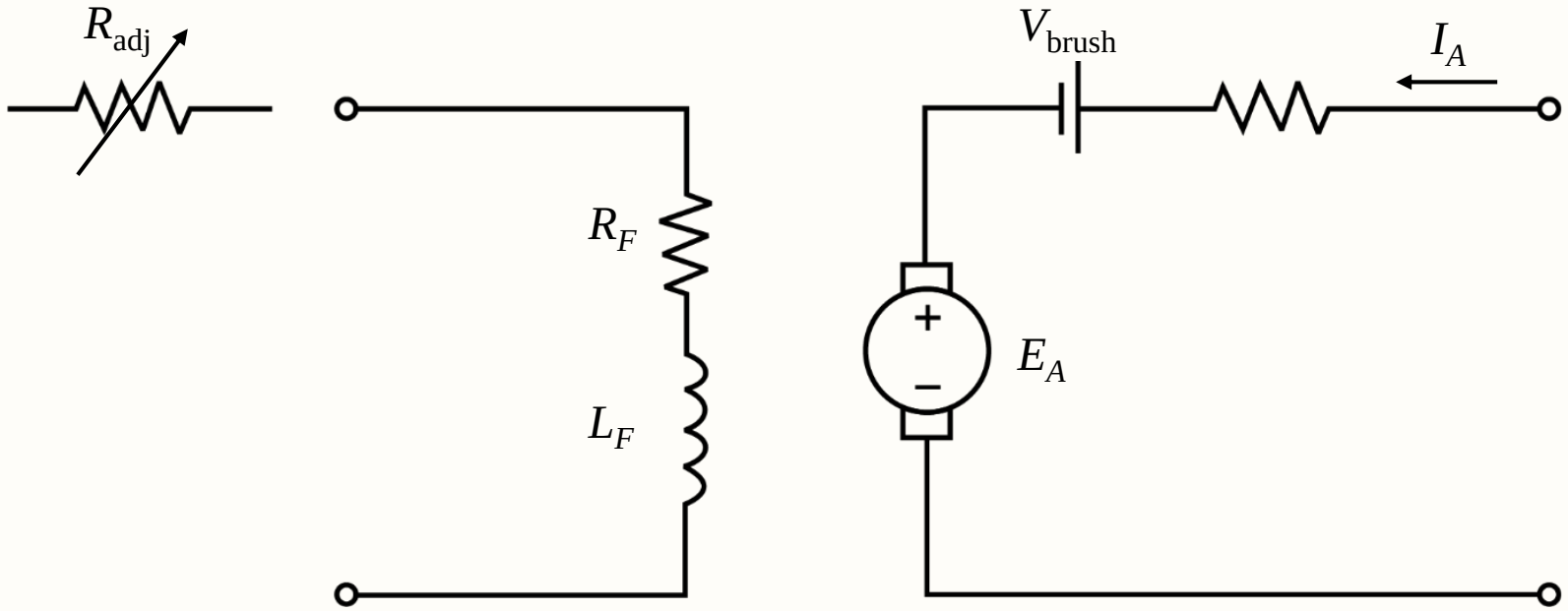
$$\text{SR} = \frac{\omega_{m,nl} - \omega_{m,fl}}{\omega_{m,fl}} \times 100\%$$

$$\Rightarrow 0.04 = \frac{n_{m,nl} - 1750}{1750} \Rightarrow n_{m,nl} = 1820 \text{ r/min}$$

In this course, the following types of DC motors will be studied in detail:

- The separately excited DC motor
- The shunt DC motor

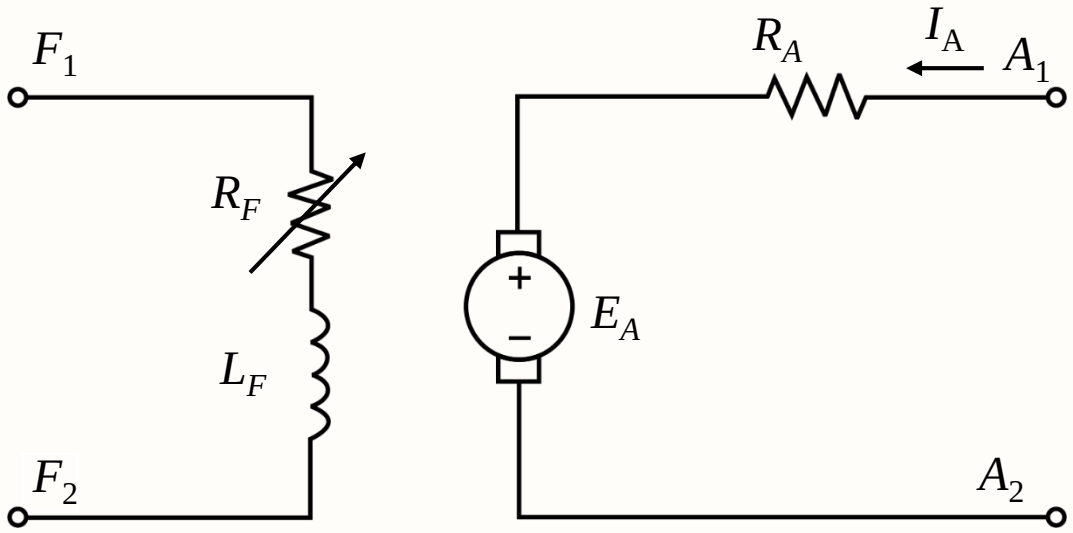
# The Equivalent Circuit of a DC Motor




Equivalent circuit of a DC motor.

- The armature (rotor) circuit is represented on the right and the field circuit on the left. The brush voltage is represented by a small battery opposing the current flow in the machine.

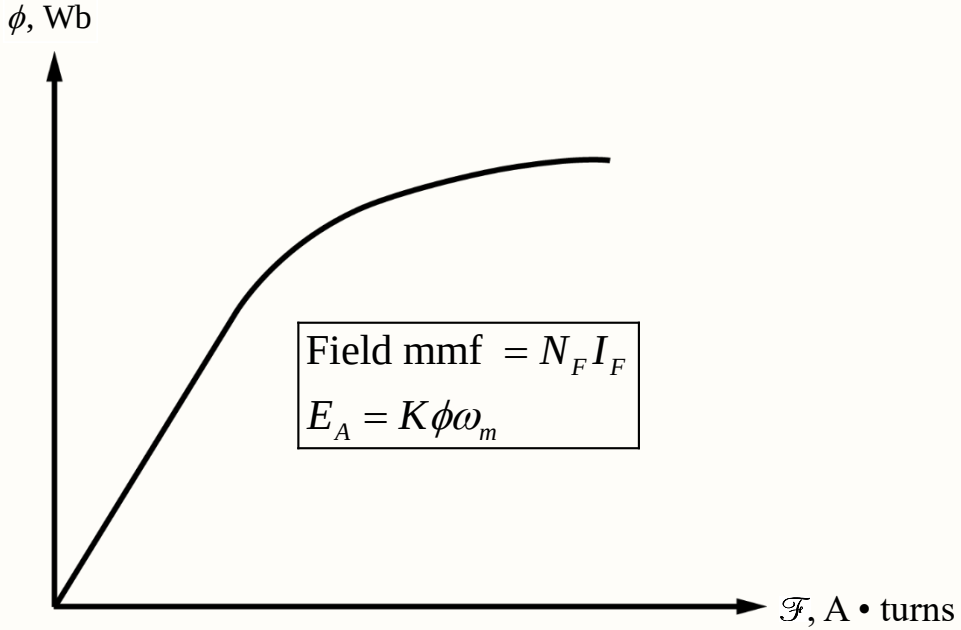
# The Equivalent Circuit of a DC Motor



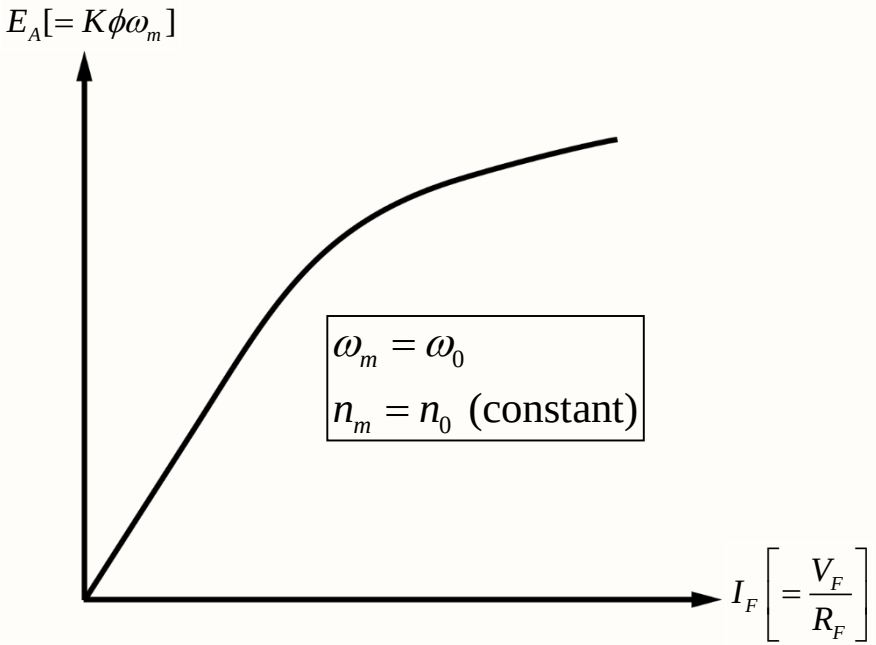

A simplified equivalent circuit of a DC motor.

- Internal generated voltage, $E_A = K\phi\omega_m$
- Induced torque, $T_{ind} = K\phi I_A$
- The induced torque $T_{ind}$ is also referred to as the developed torque $T_{dev}$.

# The Magnetisation Curve of a DC Machine




Magnetisation curve of a ferromagnetic material.



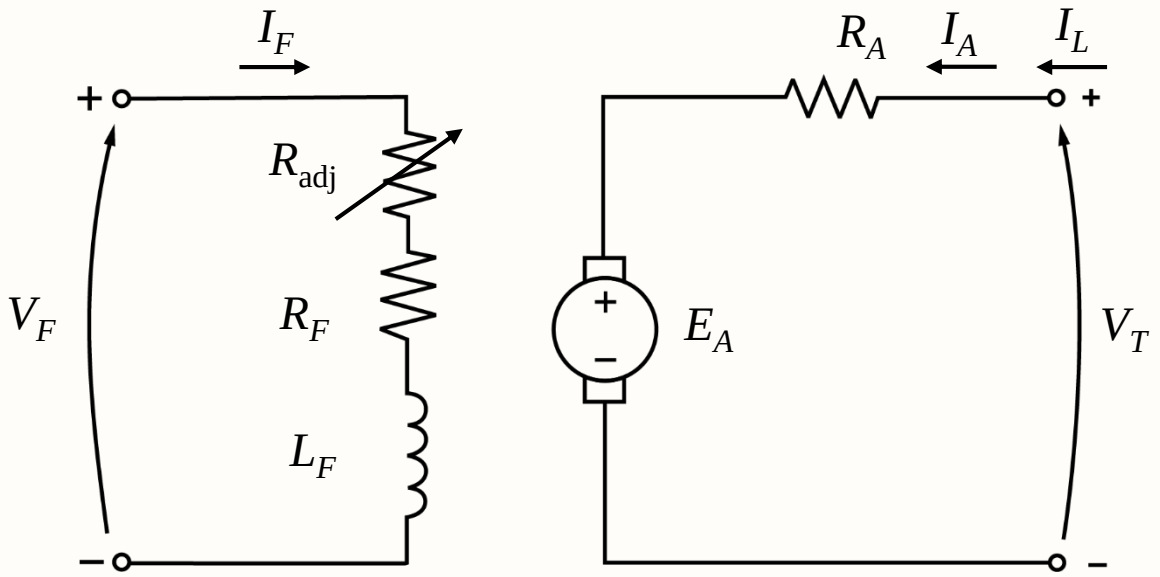

Magnetisation curve expressed as $E_A$ vs $I_F$ for fixed rotor speed $\omega_o$.

- Since $I_F$ is proportional to mmf and $E_A$ is proportional to $\phi$, we can represent the magnetisation curve as $E_A$ versus $I_F$ for a given speed $\omega_o$.

# Separately Excited DC Motor

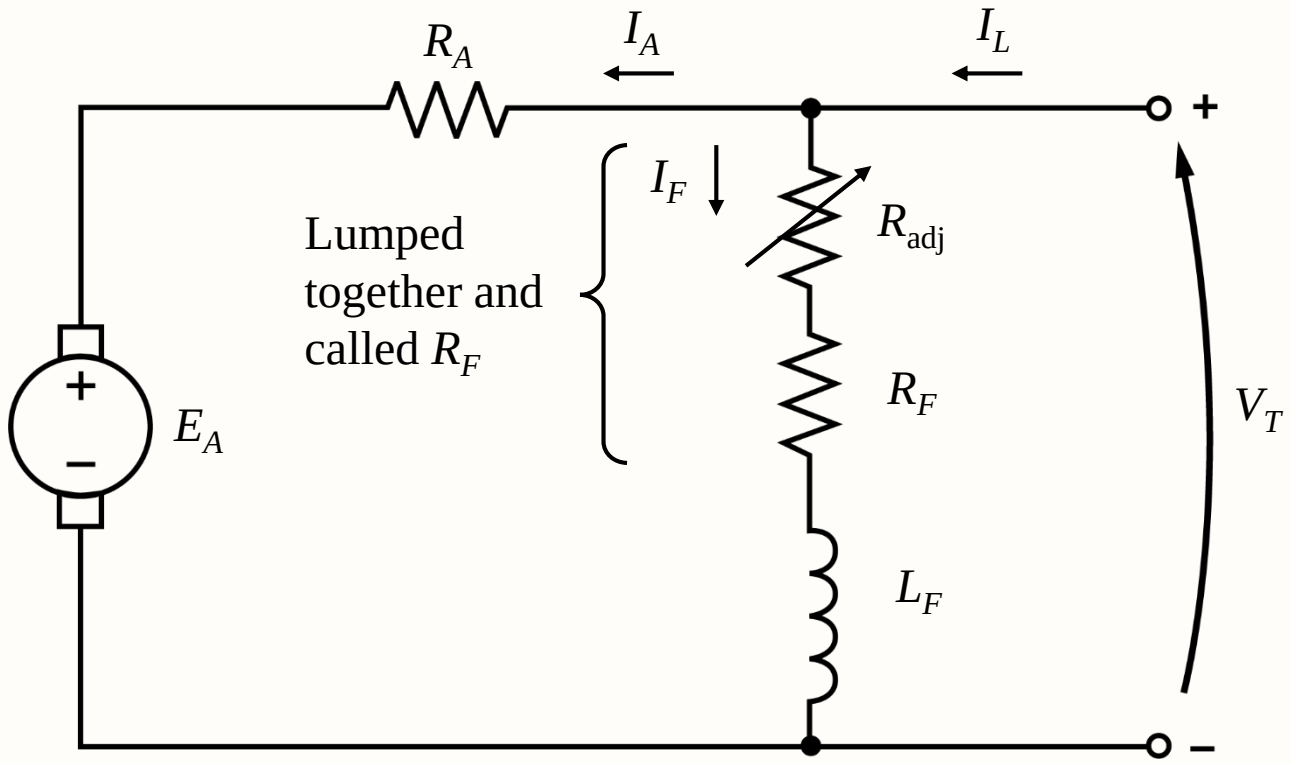
$$E_A = K\phi\omega_m,\ T_{ind} = K\phi I_A$$

$$I_F = \frac{V_F}{R_F},\ V_T = E_A + R_A I_A,\ I_L = I_A$$




Equivalent circuit.

# Shunt DC Motor




Equivalent circuit.

$$I_A = I_L - I_F$$

$$I_F = \frac{V_T}{R_F}$$

$$V_T = E_A + R_A I_A$$

$$E_A = K\phi\omega_m$$

$$T_{ind} = K\phi I_A$$

# Shunt DC Motor

- Separately excited DC motor – the field circuit is supplied from a separate constant-voltage power supply.
- Shunt DC motor – the field circuit gets its power supply directly across the armature terminals of the motor.
- When the supply voltage is constant, there is no practical difference in behaviour between the two machines.
- Unless otherwise stated, whenever the behaviour of a shunt motor is described, the separately excited motor is included too.

# The Terminal Characteristics of a Shunt DC Motor

- It is a plot of its output torque versus speed.

From $E_A = K\phi\omega_m$ and $I_A = \frac{T_{ind}}{K\phi}$,

and then substituting into $V_T = E_A + R_A I_A$

$$\Rightarrow \omega_m = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2} T_{ind}$$

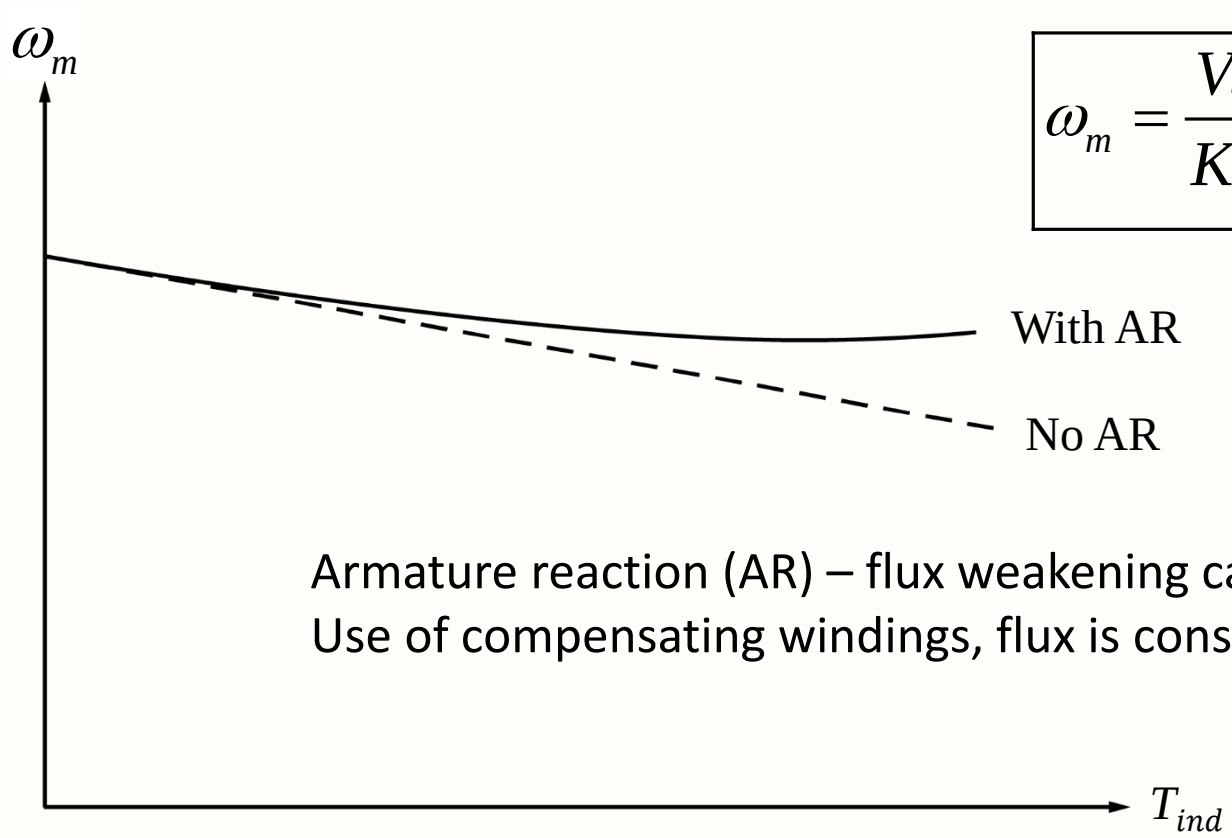
This equation is just a straight line with a negative slope.

# The Terminal Characteristics of a Shunt DC Motor

- How does a shunt DC motor respond to a load?
  - Suppose the load on motor shaft ↑.
  - Then $T_{load} > T_{ind}$ in the machine.
  - Motor will start to slow down.
  - When the motor slows down, $E_A = K\phi\omega_m \downarrow$, and so $I_A = (V_T - E_A)/R_A \uparrow$.
  - As $I_A \uparrow$, the induced torque in the motor $T_{ind} \uparrow$.
  - Finally, $T_{load} = T_{ind}$ at a lower $\omega_m$ .

# The Terminal Characteristics of a Shunt DC Motor

$$\omega_m = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2} T_{ind}$$




Armature reaction (AR) – flux weakening causes speed to increase.
Use of compensating windings, flux is constant.

Torque-speed characteristics of a shunt or separately excited DC motor with no AR, and with AR.

# Example 2

A 30-hp, 500-V, 850 r/min DC shunt motor draws a line current of 51.0 A. The resistance of the armature windings is 0.602 ohms and the combined resistance of the compensating and interpoles windings is 0.201 ohm. Its field resistance is 408.5 ohms. Calculate the internal generated voltage.

$$I_F = \frac{V_T}{R_F} = \frac{500}{408.5} = 1.224\,\text{A}$$

$$I_A = I_L - I_F = 49.776\,\text{A}$$

$$V_T = E_A + R_A I_A$$

$$E_A = V_T - I_A(0.602 + 0.201) = 460\,\text{V}$$

# Losses and Efficiency of a DC Motor

- Consider the DC shunt motor as an example:

Power input, $P_{in} = V_T I_L$

Power output, $P_{out} = T_{load} \omega_m$

$$P_{dev} = E_A I_A$$

$$P_{out} = P_{dev} - P_{rot}$$

where $P_{rot}$ are the rotational losses representing the core and mechanical losses lumped together.

Efficiency, $\eta = \frac{P_{out}}{P_{in}} \times 100\%$

Note: For separately excited motor, $\boldsymbol{P_{in} = V_T I_L + V_F I_F}$

# Determination of DC Machine Parameters

It is imperative to have access to the magnetisation curve as well as the machine parameters to analyse the performance of the machine. Run the machine with a separately excited connection and carry out the following tests:

- Open-circuit test – magnetisation curve
  - Run the DC machine as a generator at a certain speed $\omega_m$ and its terminal voltage $V_T$ is measured as the field current $I_F$ is varied.
  - Since armature circuit is open-circuited, $E_A = V_T$ .
  - As $I_F$ is varied, the generated emf $E_A$ also varies.
  - Plot $E_A$ versus $I_F$ to obtain the magnetisation curve.

# Determination of DC Machine Parameters

- Blocked-rotor test – armature and field resistances
  - The rotor is locked to inhibit rotation so that $\omega_m = 0$ and thus, $E_A = 0$.
  - Apply separate DC supplies $V_T$ and $V_F$ to the armature and field circuits, respectively.
  - Adjust till corresponding rated $I_A$ and $I_F$ flow.
  - Then $R_A = \frac{V_T}{I_A}$ and $R_F = \frac{V_F}{I_F}$

# Determination of DC Machine Parameters

- No load test – rotational losses
  - The machine is allowed to run freely at no load, without any mechanical load attached to it.
  - Both $V_T$ and $I_A$ are measured.
  - There is no output power to the machine at no-load and $E_A = V_T - I_A R_A$ where $R_A$ is found from the blocked-rotor test.
  - Then $P_{rot} = P_{dev} = E_A I_A$ can be found.

# Example 3

A shunt motor rotating at 1500 rpm is fed by a 120-V source. The line current is 51 A and the field resistance is 120 ohms. If the armature resistance is 0.1 ohm, calculate:

a) The current in the armature.

b) The internal generated voltage.

c) Mechanical power developed by motor.

d) Motor efficiency, if the rotational losses are negligible.

(Solutions →)

# Example 3 – Solutions

a) $$I_F = \frac{V_T}{R_F} = \frac{120}{120} = 1 \text{ A}$$

$$I_A = I_L - I_F = 50 \text{ A}$$

b) $$E_A = 120 - 50(0.1) = 115 \text{ V}$$

c) $$P_{dev} = E_A I_A = 5750 \text{ W}$$

d) $$P_{out} = P_{dev} - (P_{rot} = 0) = 5750 \text{ W}$$

$$\text{Eff} = \frac{P_{out}}{(P_{in} = 120 \ (51))}(100\%) = 93.95\%$$

# Example 4

A 240-V DC shunt motor has rotational losses amounting to 790 W. The field resistance is 24 ohms and the armature resistance is 0.1 ohm. If the line current is 100 A, calculate the motor efficiency.

$$P_{in} = V_T I_L = 240(100) = 24 \text{ kW}$$

$$I_F = \frac{V_T}{R_F} = \frac{240}{24} = 10 \text{ A}; \; I_A = I_L - I_F = 90 \text{ A}$$

$$E_A = 240 - 90(0.1) = 231 \text{ V}$$

$$P_{dev} = E_A I_A = 20790 \text{ W}; \; P_{out} = P_{dev} - P_{rot} = 20 \text{ kW}$$

$$\text{Eff} = \frac{P_{out}}{P_{in}}(100\%) = 83.33\%$$

# Example 5

A 50-hp, 250-V DC shunt motor has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06 ohms. Its total field circuit resistance is 50 ohms. Assume that the no-load speed is 1200 r/min and, at no load, the armature current is negligible, so that $E_A \simeq V_T = 250\ \mathrm{V}$. There are 1200-turns per pole on the field winding.

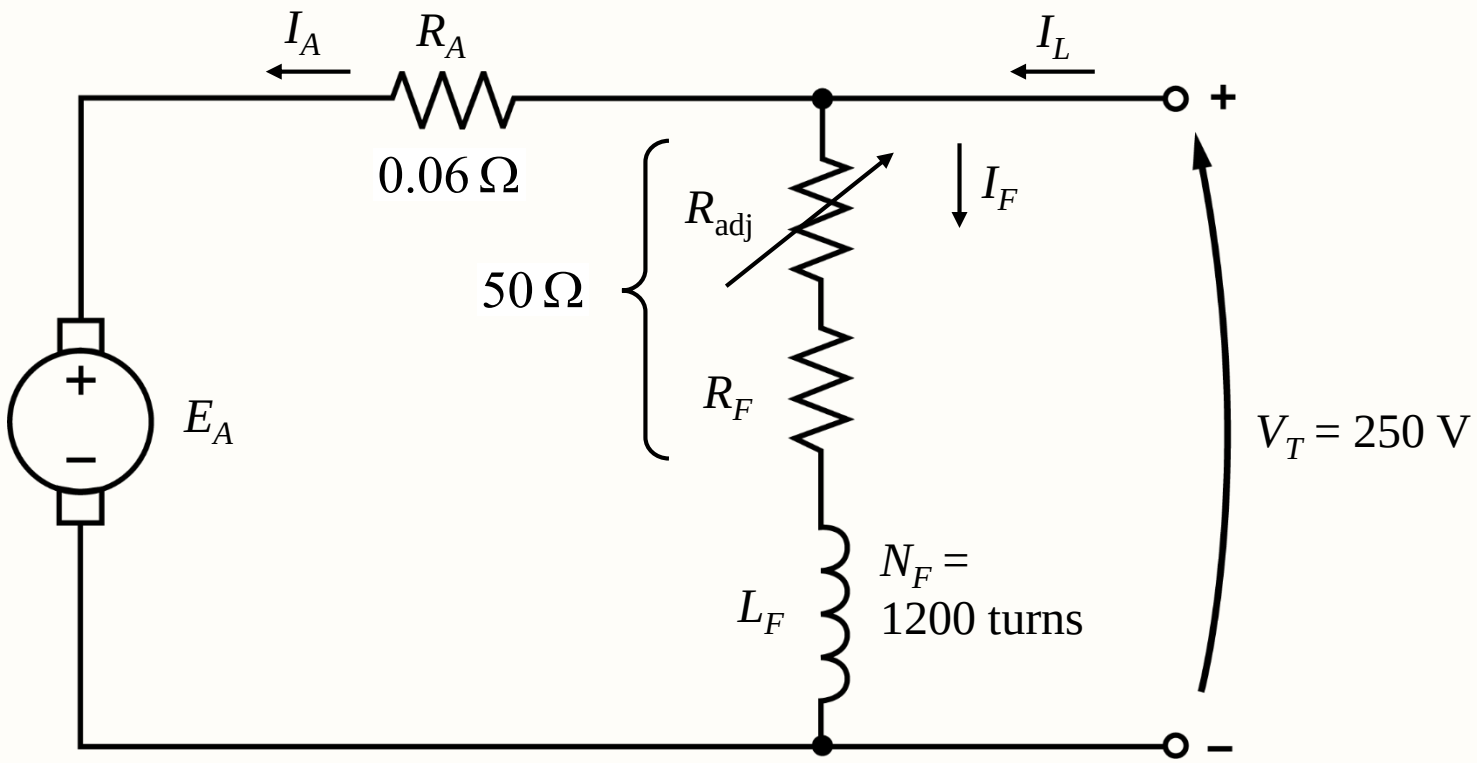
a) Find the speed of this motor when its input current is 100A.

b) Find the speed of this motor when its input current is 200A.

c) Find the speed of this motor when its input current is 300A.

# Example 5

d) Plot the torque-speed characteristics of this motor.




(Solutions →)

# Example 5 – Solutions

Since the field current is constant, and since there are no armature reaction, the flux is constant.

From $E_A = K\phi\omega_m = \hat{K}\phi n_m,\ n_m$ in rev/min

$$\Rightarrow \frac{E_{A2}}{E_{A1}} = \frac{\hat{K}\phi n_{m2}}{\hat{K}\phi n_{m1}} = \frac{n_{m2}}{n_{m1}}$$

a) At no load, $E_{A1} \simeq V_T = 250\ \text{V},\ n_{m1} = 1200\ \text{rev/min}$

If $I_L = 100\ \text{A},\ I_A = I_L - I_F = 95\ \text{A}$

$$E_{A2} = V_T - I_A R_A = 244.3\ \text{V}$$

$$\Rightarrow n_{m2} = \frac{E_{A2}}{E_{A1}}(1200) = 1173\ \text{rev/min}$$

# Example 5 – Solutions

b) If $I_L = 200$ A, $I_A = I_L - I_F = 195$ A

$$E_{A3} = V_T - I_A R_A = 238.3 \text{ V},$$

$$\Rightarrow n_{m3} = \frac{E_{A3}}{E_{A1}}(1200) = 1144 \text{ rev/min}$$

c) If $I_L = 300$ A, $I_A = I_L - I_F = 295$ A

$$E_{A4} = V_T - I_A R_A = 232.3 \text{ V},$$

$$\Rightarrow n_{m4} = \frac{E_{A4}}{E_{A1}}(1200) = 1115 \text{ rev/min}$$

# Example 5 – Solutions

d) The electrical power that is converted to mechanical power,

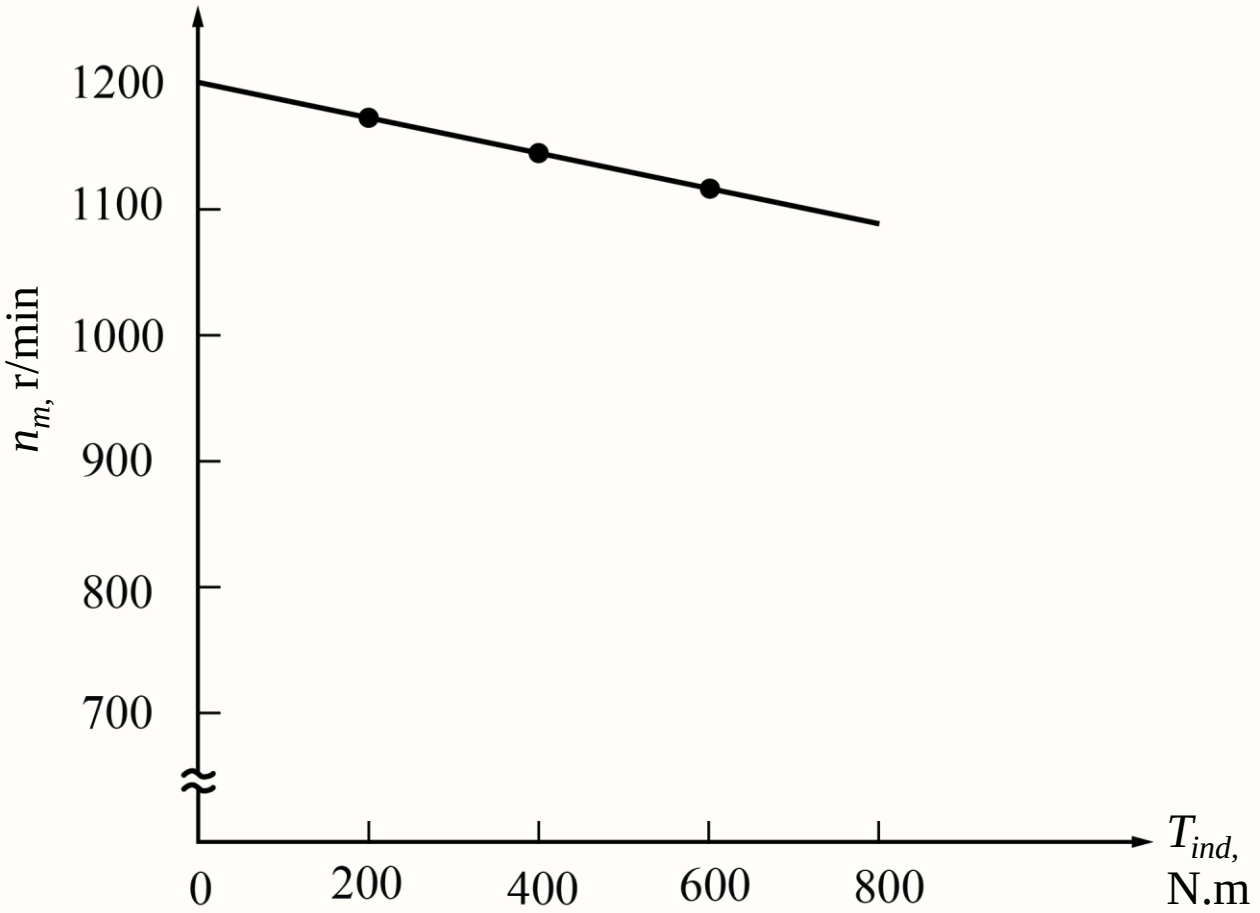
$$P_{conv} = P_{dev} = E_A I_A = T_{ind}\omega_m \Rightarrow T_{ind} = \frac{E_A I_A}{\omega_m}$$

The torque developed or torque induced when $I_L = 100$ A is

$$T_{ind} = \frac{244.3(95)}{(1173)(2\pi/60)} = 189 \text{ N.m.}$$

Similarly, the induced torque when $I_L = 200$ A is 388.2 N.m, and when $I_L = 300$ A is 586.9 N.m.

# Determination of DC Machine Parameters



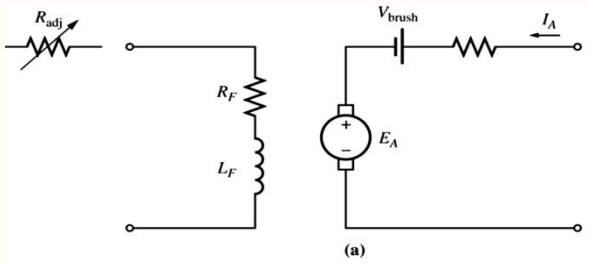
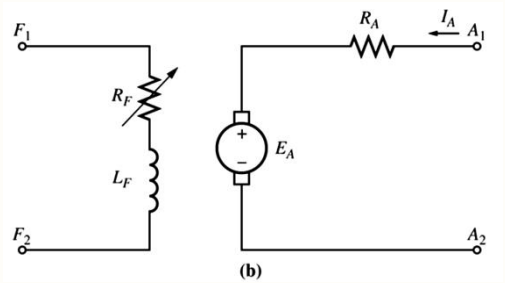
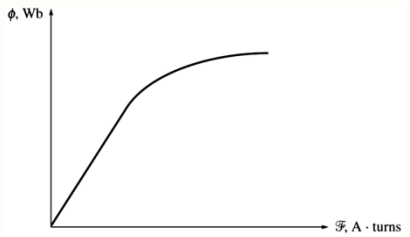

Torque-speed characteristics of the motor in the example.

# Summary

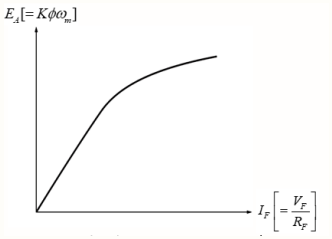
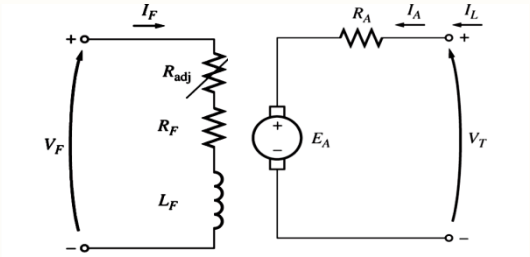
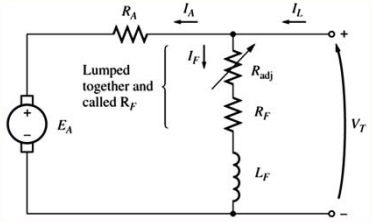
In this lecture, you have learnt:

- The equivalent circuit of a DC motor.
- The torque-speed characteristics of a shunt or separately excited DC motor.
- The power flows and analysis of the motor from the equivalent circuit.

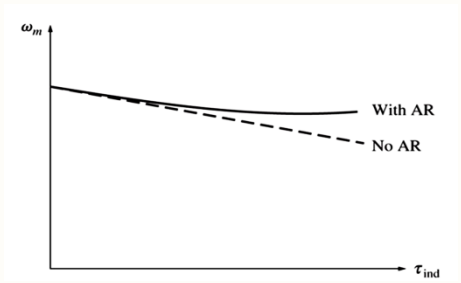
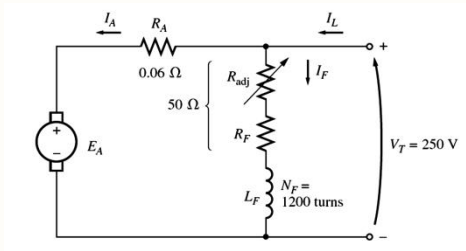
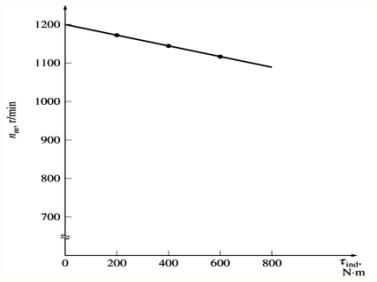
# References

| No. | Slide No. | Image | Reference |
|---|---|---|---|
| 1 | 6 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 467), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |
| 2 | 7 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 467), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |
| 3 | 8 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 468), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |

# References

| No. | Slide No. | Image | Reference |
|---|---|---|---|
| 4 | 8 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 469), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |
| 5 | 9 |  | Adapted from *Electric Machinery Fundamentals, 5th ed.*, (p. 470), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Adapted with permission. |
| 6 | 10 |  | Adapted from *Electric Machinery Fundamentals, 5th ed.*, (p. 470), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Adapted with permission. |

# References

| No. | Slide No. | Image | Reference |
|---|---|---|---|
| 7 | 14 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 472), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |
| 8 | 24 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 473), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |
| 9 | 28 |  | Reprinted from *Electric Machinery Fundamentals, 5th ed.*, (p. 475), by S. J. Chapman, 2012, New York, NY: McGraw-Hill. Copyright 2012 by The McGraw-Hill Companies, Inc. Reprinted with permission. |